3A01 Reflection of Light

3.1.1 Laws of reflection

The incident ray, reflected ray and normal lie in the same plane, and the angle of reflection equals the angle of incidence. Both angles are measured from the normal, not the mirror surface. For a curved surface, use the local normal at the incidence point. Draw this first, then draw the reflected ray at the matching angle.

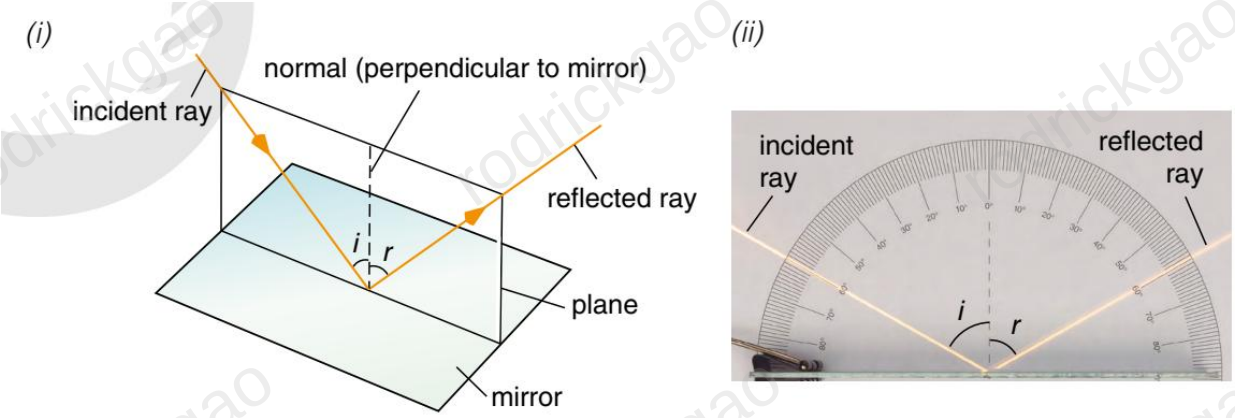


Fig 1.2c *The laws of reflection.*

The incident ray, reflected ray and normal share one plane. Both i and r are measured from the normal, giving i = r.

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3.1.2 Images in a plane mirror

A plane mirror forms an upright, same-sized, laterally inverted virtual image. Its perpendicular distance behind the mirror equals the object's perpendicular distance in front. Reflected rays do not actually pass through the image point; only their backward extensions meet there, so the image cannot be projected onto a screen.



Fig 1.2j *The image formed by a mirror is laterally inverted.*

The original photograph shows lateral inversion. The virtual image appears behind the mirror, with the same size and perpendicular distance as the object.

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